Blepharoplasty, Canthoplasty, and Related Procedures

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Link to Codes

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Clinical Indications

- Blepharoplasty, canthoplasty, or related procedures are indicated when **ALL** of the following are present(1)(2)(3):
  - Eyelid problem requiring treatment, as indicated by **1 or more** of the following:
    - Ectropion,[A] with evidence of symptomatic corneal exposure (eg, excessive drying, tearing, irritation, foreign body sensation, keratitis, corneal ulcer)(9)(11)(12)(13)
    - Entropion,[B] when local measures fail to control symptoms such as eye pain or corneal irritation(9)(15)(17)
    - Exposure keratitis due to **1 or more** of the following(12):
      - Eyelid laxity
      - Inability to properly close eye due to Bell palsy or other disorder(11)(19)(20)
      - Postoperative complication (eg, absence of part of eyelid from previous surgery)(18)
    - Ptosis of lid[C] or dermatochalasis,[D] as indicated by **1 or more** of the following(4)(23):
      - Congenital ptosis with amblyopia[E](29)(30)(31)(32)
      - Margin reflex distance 1 (MRD_1) less than or equal to 2 mm[F] in central gaze[G]
      - Margin reflex distance 1 (MRD_1) less than or equal to 2 mm[F] in down gaze[H] with impairment of reading
      - Visual field testing shows superior visual field loss[I] of 12 degrees of vision or 24% impairment.[J](36)
  - Preoperative ophthalmologic examination[K] has been performed.(4)(37)(38)(39)

Alternatives

- Alternatives include eye lubricant therapy.(15)(20)

Evidence Summary

Background

Blepharoplasty is a general term for plastic surgery on the eyelids.(4)(5)(6) **(EG 2)** It can involve either the upper or lower lid, as well as their medial and lateral margins; it may also involve canthoplasty, which is plastic surgery of the medial and/or lateral canthus (the angle formed by the meeting of the upper and lower eyelids at either side of the eye).(7)(8) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 31 published peer reviewed articles, 1 specialty society or other evidence-based guideline, 1 Cochrane systematic review, and 5 book sections.

For ectropion, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A retrospective comparative case series of 67 eyes with involutional ectropion concluded that the lateral tarsal strip procedure without a medial spindle (tarsoconjunctival diamond excision) was effective in a large subset of patients who had a good result on preoperative lateral pinch and twist test.(14) **(EG 2)**

For entropion, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Mild symptoms of entropion can sometimes be treated with lubrication with or without electroepilation (the electrical permanent destruction of the eyelash follicle), but many patients require surgical treatment to reconstruct the eyelid margin.(15)(17) **(EG 1)**

For exposure keratitis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A study of 7 patients with adult-onset exposure keratitis after childhood ptosis repair showed that these patients were successfully treated with lubricating drops with or without further surgery.(18) **(EG 2)**

For ptosis or dermatochalasis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Upper eyelid ptosis is evaluated by the relation of the eyelid margin to the superior limbus of the iris in central gaze.(24)(25) **(EG 2)** The normal position of the upper lid margin covers 2 to 3 mm of the superior limbus but does not cover any aspect of the pupil unless ptosis is present.(25)(26) **(EG 2)** Ptosis can be repaired with a variety of techniques, depending upon the adequacy of levator muscle function.(4)(25)(26)(27) **(EG 2)** A prospective cohort study of 466 ptosis repairs in Asian patients found improvement in margin reflex distance 1 (MRD_1) in 93.8% of patients; 70.5% had a postoperative MRD_1 of greater than or equal to 2 mm.(28) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(40); 42 CFR 422.101(41)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(42); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(43); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(44)

Medicare Coverage Determinations: Medicare Coverage Database(45)

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Footnotes

[A] Ectropion is an outward turning of the eyelid (eversion) so that the lid margin does not rest against the eyeball; it can cause corneal exposure, with excessive drying, tearing, irritation, and ulceration.(9)(10) [A in Context Link 1]

[B] Entropion is an inward turning of the upper or lower eyelid so that the lid margin rests against and rubs the eyeball, which can lead to trichiasis (eyelashes rubbing on the eyeball), permanent corneal scarring, and pain.(15)(16) [B in Context Link 1]

[C] Ptosis (blepharoptosis) is a downward displacement of the upper eyelid margin, which can result in partial or complete obstruction of the patient's visual field.(4)(7)(21) [C in Context Link 1]

[D] Dermatochalasis is redundant or excess eyelid skin and can also involve underlying muscle, connective tissue, and fat.(4)(21)(22) [D in Context Link 1]

[E] Amblyopia is a disorder of visual acuity that is not entirely attributable to structural abnormalities; it is caused by early childhood visual experiences. Amblyopia is categorized according to the causative disorder: strabismic, refractive, visual deprivation, or occlusion (eg, reverse).(31)(33) [E in Context Link 1]

[F] Margin reflex distance 1 (MRD_1) is a measurement of the distance from the central corneal light reflex to the upper eyelid margin. An MRD_1 of 2 mm in primary gaze has been shown to correlate with superior visual field impairment of 24% to 30% and superior visual field loss of 12 to 15 degrees.(4) [F in Context Link 1, 2]

[G] Central gaze is the position in which both of the patient's eyes are fixated on a distant object straight in front of the patient.(34) [G in Context Link 1]

[H] Down gaze is the position in which both of the patient's eyes are fixated on an object 40 degrees downward from central gaze, such as when reading.(34)(35) [H in Context Link 1]

[I] Superior visual field loss is associated with functional deficits.(4) [I in Context Link 1]

[J] Visual field testing maps the central and peripheral vision of the individual eyes separately, often using automated perimetry equipment. A normal unobstructed visual field extends 50 to 60 degrees superiorly.(4) [J in Context Link 1]

[K] A preoperative ophthalmologic examination should include a complete periocular examination, including the brow, lid, cheek, and ocular surface, and a complete ocular examination, including documentation of vision and evaluation of pupil, extraocular motility, and corneal status via slit-lamp examination.(37) [K in Context Link 1]

Codes

CPT®: 15820, 15821, 15822, 15823, 67901, 67902, 67903, 67904, 67906, 67908, 67914, 67915, 67916, 67917, 67921, 67922, 67923, 67924, 67950 [Hide]

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